

A Lesion-Wise Cascade for False-Positive Reduction and Lesion-Level Brain Tumor Typing in Multi-Sequence MRI

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1 MOTIVATION

- Missed lesions in SRS workflows may forfeit treatment windows.
- Sensitivity-first detectors reduce misses but create a heavy false-positive burden.
- Brain-mass review requires imaging-based differential diagnosis beyond detection.
- Lesion-wise typing preserves intra-patient heterogeneity for expert review.

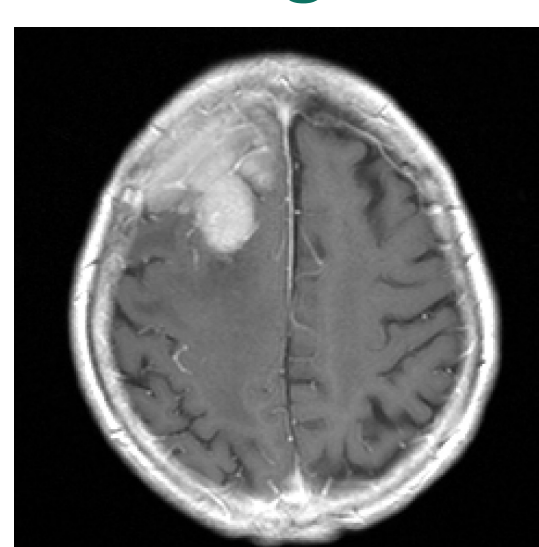


Goal: Suppress FPs without sacrificing sensitivity and assign each retained lesion one of five tumor subtypes.

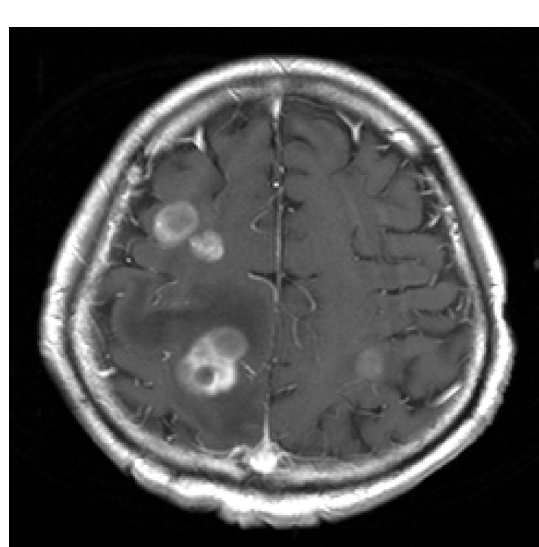
Neuroma



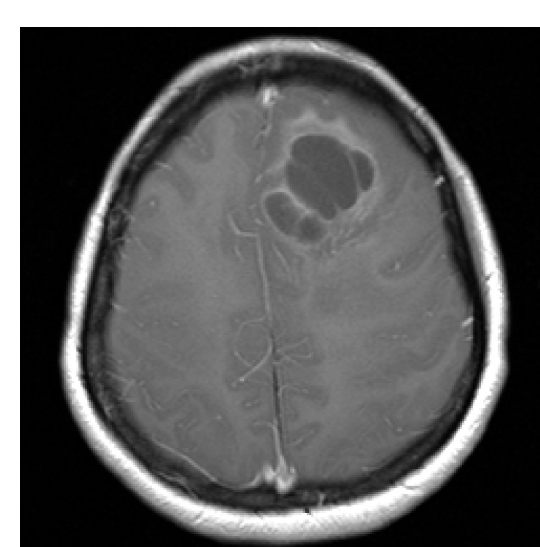
Meningioma



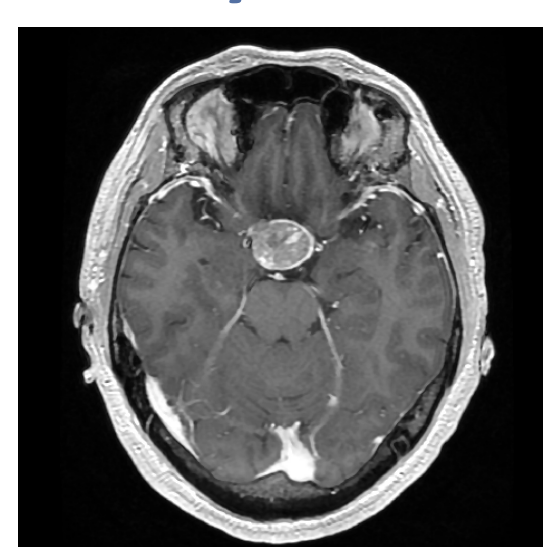
Metastasis



Glioma

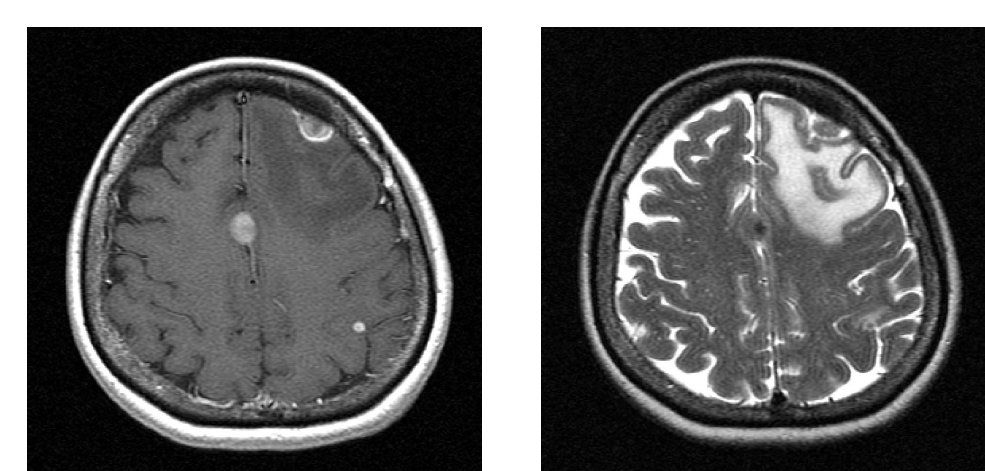


Pituitary adenoma



2 METHOD: THREE-STAGE CASCADE PIPELINE

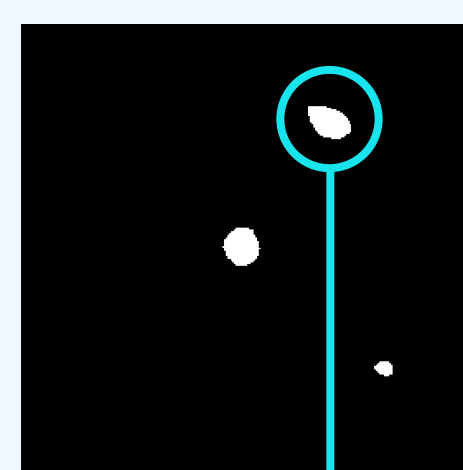
Input: Co-registered T1C + T2 MRI



STAGE 0

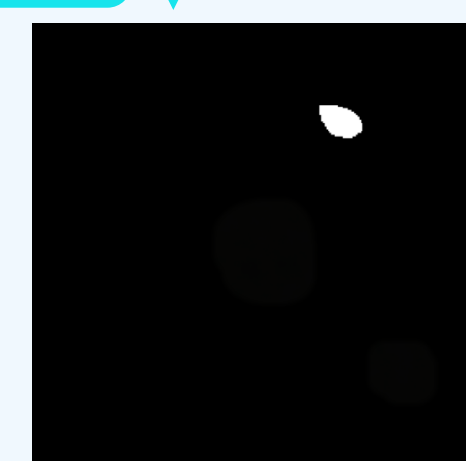
Sensitivity-first Candidate Generation 3D nnU-Net v2

- 3D volumetric detector
- High sensitivity candidate mask



Lesion-Wise Patch Construction

- Full in-plane FOV resized to 128×128
- Candidate-centered 15-slice axial stack
- Channels: T1C, T2, predicted mask
- Robust z-score in 5 mm dilated ROI



one candidate = one patch

STAGE 1

Mask-gated Binary TP/FP Classifier Kinetics-400-pretrained R(2+1)D-18

- CE warm-up → Asymmetric focal loss
- Balanced sampling



STAGE 2

Multi-scale Mask-gated Final Classifier (FP + 5-class typing) Kinetics-400-pretrained R(2+1)D-18

- Binary CE (FP head) + Focal CE (subtype head)
- Prevalence-aware sampling
- Parallel dilated conv. In 3 scales



Neuroma

Meningioma

Metastasis

Glioma

Pituitary adenoma

4 RESULTS OVERVIEW (vs Stage-0)



90.13%

Final Sensitivity

93.72% → 90.13%

↓ **3.59%**



0.41

Final FP/scan

1.53 → 0.41

↓ **73.2%**

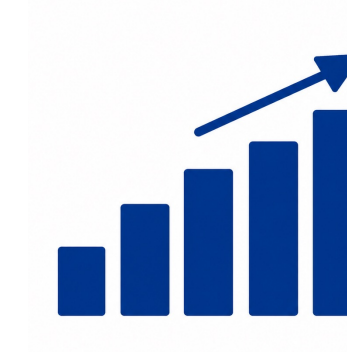


66.34%

Final Precision

35.42% → 66.34%

↑ **30.92%**



89.55%

5-class Accuracy

on retained lesions

Cascade Performance

Variant	Sensitivity (%)	Precision (%)	FP / scan	Rem. TP	Rem. FP
Stage-0 baseline	93.72	35.42	1.53	—	—
+ Stage-1 only	91.03	62.65	0.49	6	260
+ Full cascade	90.13	66.34	0.41	8	279

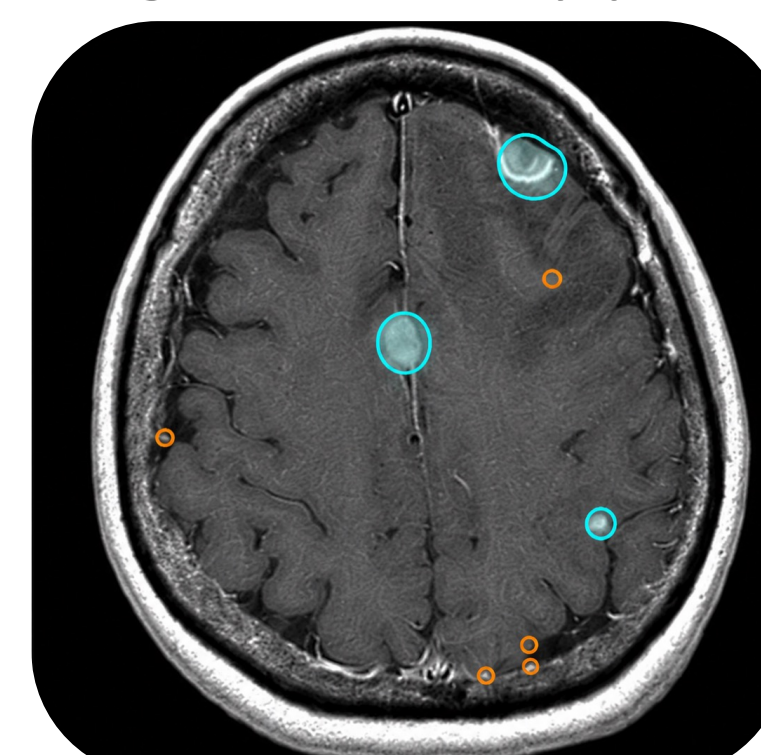
Rem. = Removed counts relative to Stage-0 baseline.

Cascade Progression

● True Positive (TP) ● False Positive (FP)

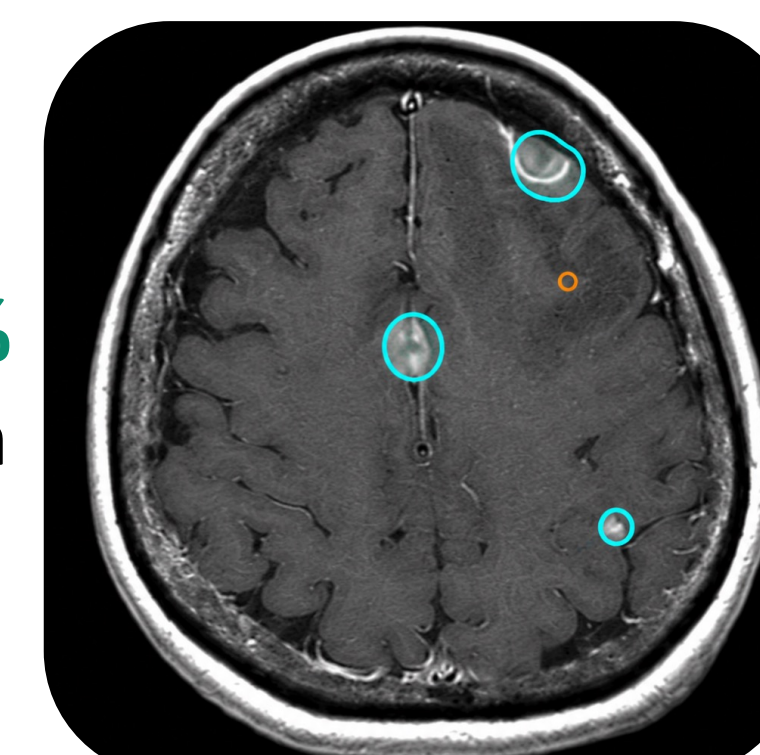
Stage-0 Baseline

Raw candidates: TP + FP
High sensitivity pool



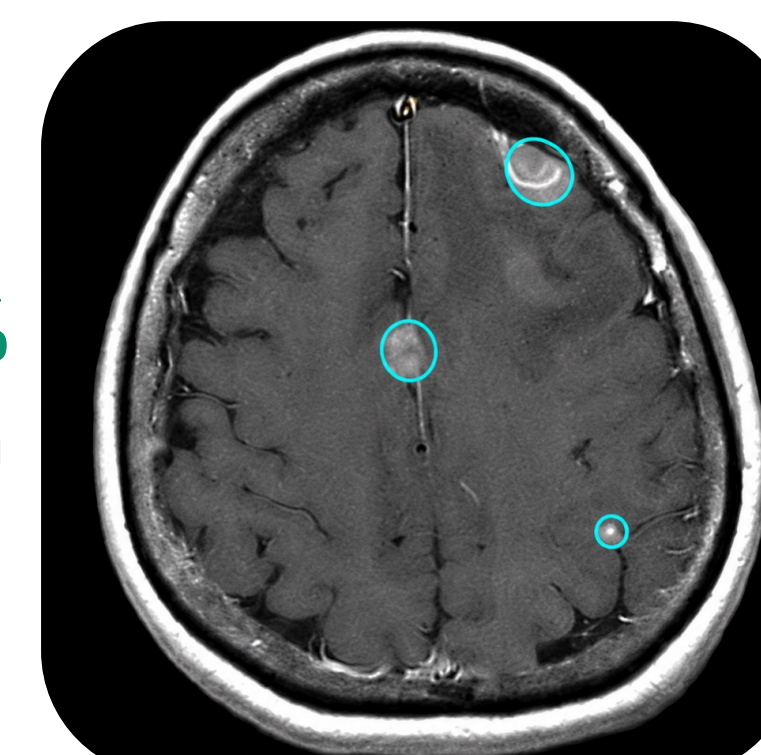
Stage-1 Only

Obvious FPs removed
Most FP reduction



Full Cascade

Residual FPs removed +
subtype assigned



-68.0%
FP/scan

-16.3%
FP/scan

Subtype Classification (n=201 lesions, Acc. = 89.55%)

		Predicted					Per-Class Sensitivity (%)
		Neu	Men	Met	Gli	Pit	
Ground Truth	Neuroma	16	0	1	1	0	88.9%
	Meningioma	1	30	3	3	0	81.1%
	Metastasis	0	2	107	10	0	89.9%
	Glioma	0	0	0	16	0	100%
	Pituitary	0	0	0	0	11	100%

3 DATA SUMMARY

Total: 12,437 dual-centre T1C/T2 MRI scans

Development cohort

- 12,188 scans
→ 20,305 Stage-0 patches
- 12,342 TP / 7,963 FP
80/20 case-wise split

Held-out test cohort

- 249 scans, 223 lesions
→ 590 candidates
- 209 TP / 381 FP
14 fixed Stage-0 FNs

CORRESPONDENCE

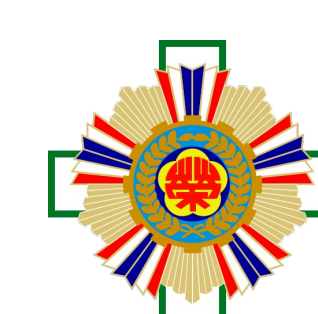
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5 CONCLUSIONS

- Stage-1 delivers most FP reduction (−260 FPs, −1.04 FP/scan; only −2.69% sensitivity).
- Stage-2 refines residual FPs (−19 more) and enables 5-class subtype classification (89.55% accuracy).
- Full cascade: 0.41 FP/scan, 90.13% sensitivity — 73.2% FP/scan reduction vs. raw candidates.
- Lesion-wise formulation avoids patient-level label aggregation, preserving subtype heterogeneity.
- Outlook: extend to longitudinal SRS surveillance; explore transformer backbones for encoder-level multi-scale attention.

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